EECS 484, Winter 2026 - Database Management Systems

DBMS Overview

Chapter 1

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What is a DBMS?

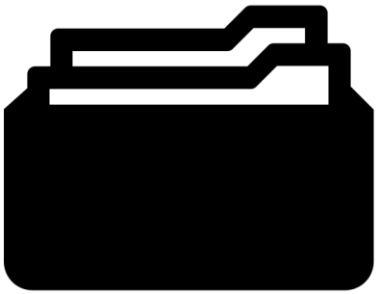
- Database: Large, integrated collection of data
- DBMS = Database Management System
  = a software package designed to store and manage databases
- Examples:
  - Oracle, MS SQL server, DB2, MySQL, Postgres
- **Goal**: Provide an abstraction layer for storing and accessing data
  - Why abstraction? For efficiency and convenience

DBMS: Motivation

- Consider the following scenario:
  - Students taking classes, and obtaining grades
  - Q1: Find grade in class X
  - Q2: Find my GPA
  - …

Old-time Solution: Sorted Student Folders

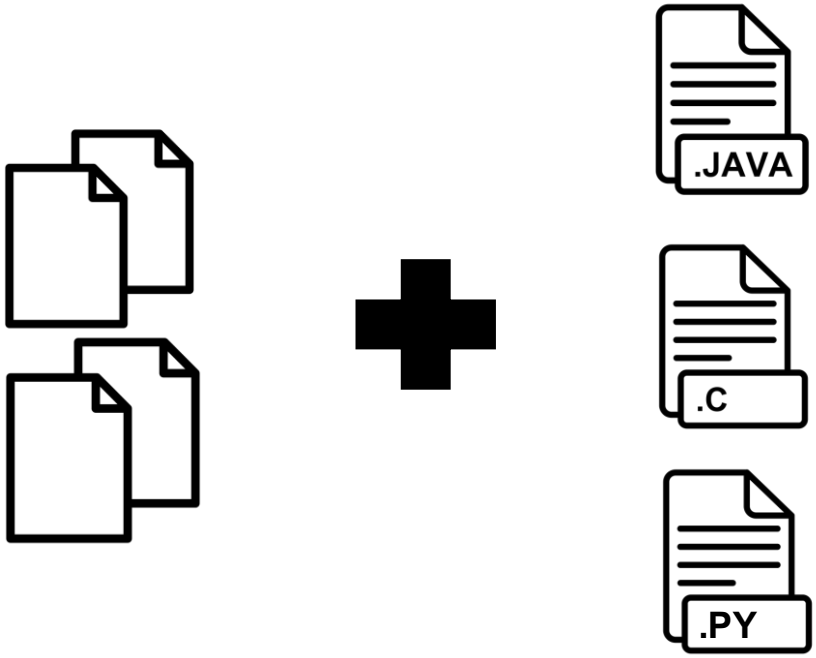
- Advantages?
- Disadvantages?



Ideas?

Other Solution: Flat Files

- Access?
  - using programs in C, Java, Python etc.
- Layout for the student records?



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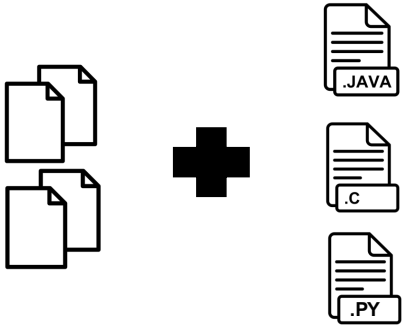

CSV:

Brown, Lisa, lbrown, 484, A+, 482, D
Smith, Bart, bsmith
Tompson, Mary, mtom, 281, B+, 484, A
…
…

Other Solution: Flat Files

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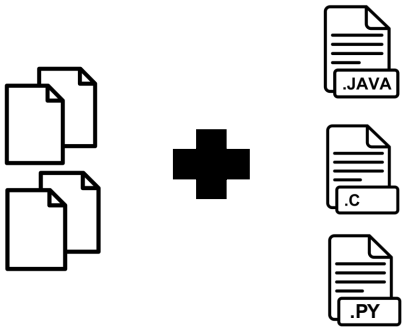

Multiple files:

Brown, Lisa, lbrown
Smith, Bart, bsmith
Tompson, Mary, mtom
…
…

lbrown, 484, A+
lbrown, 482, D
mtom, 281, B+
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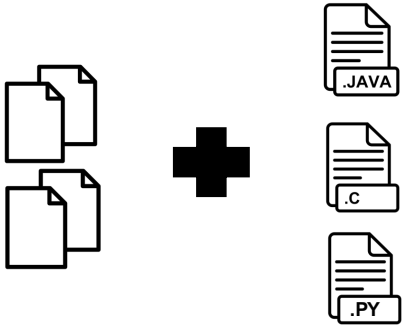
…

Other Solution: Flat Files

- Problems?



Why use a DBMS?

- It solves ALL problems with file-based data mngt.

But How?

Why use a DBMS?

- It solves ALL problems with file-based data mngt.
  - Reduces application dev time
  - Allows for efficient storage and access
  - Shields apps from changes to internal representation and storage format
  - Protects data integrity and security
  - Enables concurrent access & recovery from crashes

But HOW???

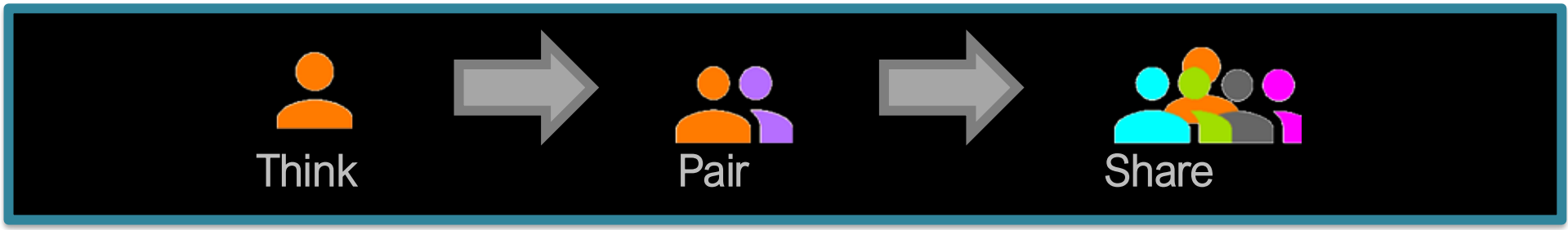
Declarative vs. Imperative Languages

- Declarative: Express **what** your goal is
  - Return all students who have a GPA higher than their last year's GPA

- Imperative: Express **how** to achieve your goal
  - Read this file,
  - For each record, add up their grades until last yr
  - Record last year's GPA in an array
  - ...

CYU: Declarative Interfaces

- Pros?
- Cons?



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Relational and Other Data Models

- **DBMS using the relational DM** ('70s-now)
  - IBM DB2
  - Informix
  - Oracle
  - Sybase
  - Microsoft Access
  - Tandem
  - Teradata
  - …
- **Other data models**
  - Hierarchical (mid '60s-'70s)
    - IBM IMS
  - Network ('70s)
    - IDMS, IDS
  - Object-oriented (~'90s)
    - ObjectStore
  - Object-relational (relational model + object DB concepts)
    - Oracle
  - …

Relational (Data) Model

- The most widely-used model today
- **Data model** = a collection of concepts for describing data
  - A collection of relations
  - Relation = set of records – think of it as a table with rows and columns

Students

sid	name	login	age
13	Lisa	lsimp	40
41	Bart	bart	20

Courses

cid	cname	cred.
E-484	EECS484	4
E-482	EECS584	4

Enrolled

sid	cid	grade
41	E-484	A+
13	E-482	C-

Relational (Data) Model

- **Schema** = a description of data in terms of a data model
  - Every relation has a schema
  - Specifies the name of the relation, the name and type of the columns (or fields or attributes)
  - Each row also called a tuple or a record

Students(sid:`string`, name:`string`, login:`string`, age:`integer`)
Courses(cid:`string`, cname:`string`, credits:`integer`)
Enrolled(sid:`string`, cid:`string`, grade:`string`)

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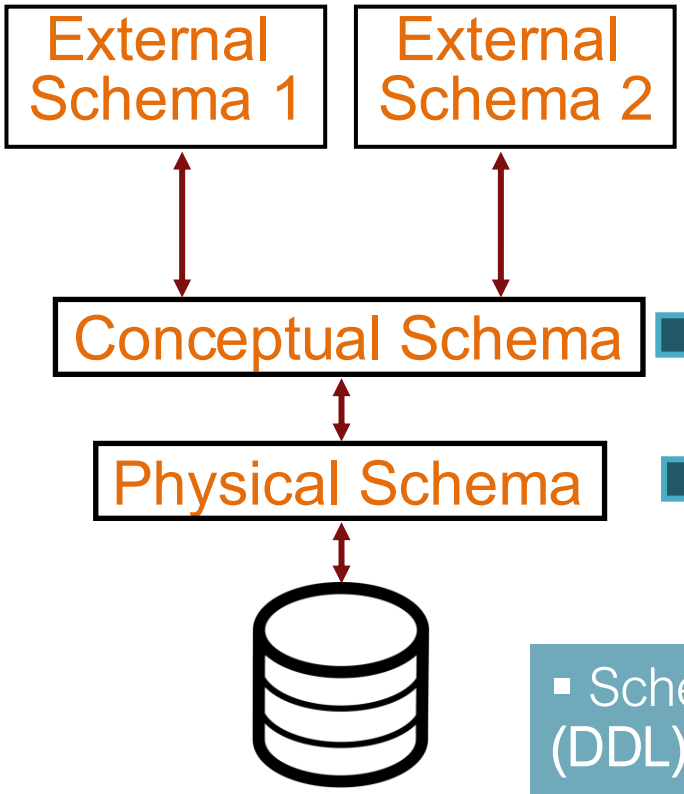
Levels of Abstraction



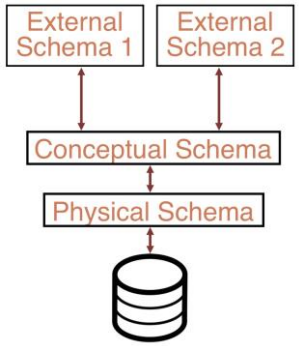
describe how users see the data, views

defines the logical structure in terms of the DBMS's data model

describes the files and indexes used

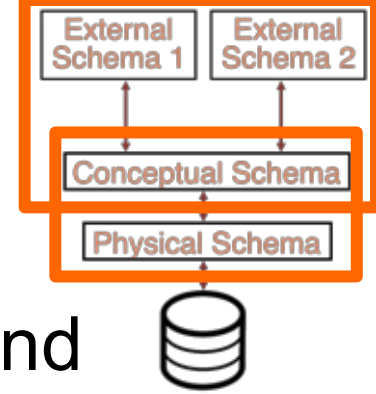
- Schemas are defined using **Data Definition Language (DDL) – e.g.,** `CREATE TABLE X (…)`
- Data is modified/queried using **Data Manipulation Language (DML) – e.g.,** `SELECT FROM X WHERE …`

Example



- Conceptual schema (1):
  - Students(sid:`string`, name:`string`, login:`string`, age:`integer`)
  - Courses(cid:`string`, cname:`string`, credits:`integer`)
  - Enrolled(sid:`string`, cid:`string`, grade:`string`)
- Physical schema (1):
  - Relations stored as unordered files.
  - Index on first column of Students.
- External Schema (≥ 1):
  - View: Course_info(cid:`string`, enrollment:`integer`)
  - View: Class_rank(sid:`string`, gpa:`real`, rank:`integer`)

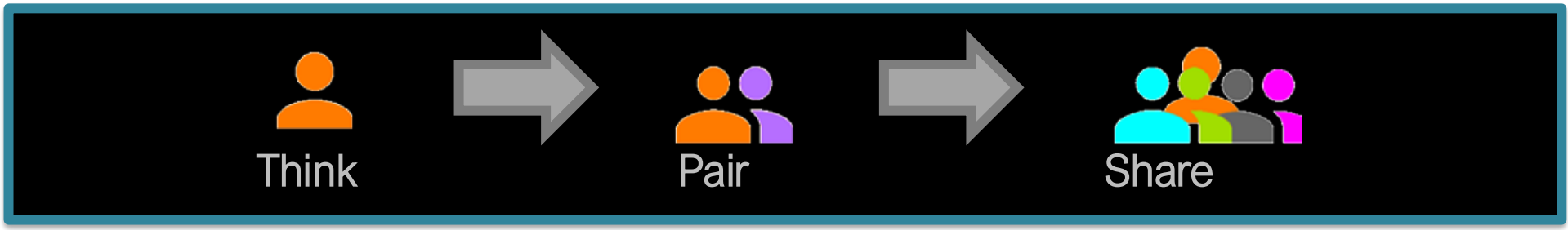
Data Independence



- Applications insulated from data format and storage details
- Logical data independence: Protection from changes in *logical* structure of data
  - External schemas
- Physical data independence: Protection from changes in *physical* structure of data
  - Conceptual schemas

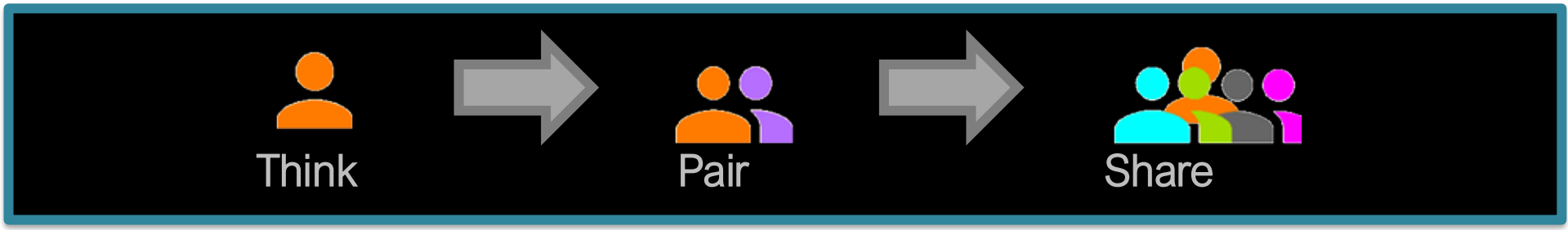
CYU

- Which of these are more suitable for storing in a DBMS rather than files in an OS?
  - (a) Grades for students at the university
  - (b) Source code for a program
  - (c) Contents of a textbook



CYU

- While UM uses a single relational table with information about both your grades and number of credits, Canvas only allows you to see your letter grades. Does Canvas represent:
  a) An external schema?
  b) A conceptual schema?
  c) A physical schema?



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  - Transaction support
    - Enables concurrent access & recovery from crashes

Who uses a DBMS?

- Everyone!
  - Your bank
  - Your university
  - Your coffee shop
  - Your favorite hotel
  - Your favorite website
  - Your phone
  - Your government
- How many databases have you used so far today?

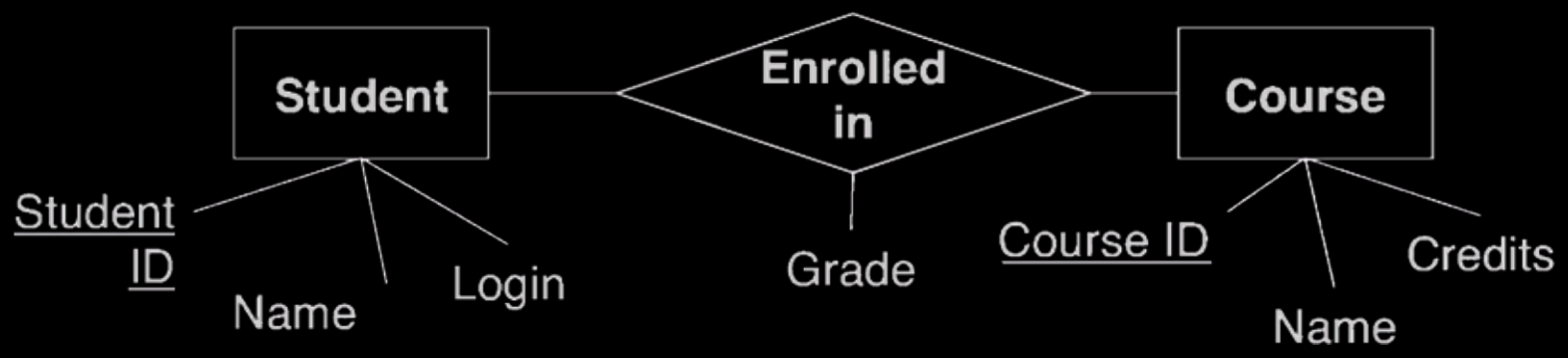
Database Administrator (DBA)

- People in charge of running a DBMSs
  - Designs logical /physical schemas
  - Handles security and authorization
  - Data availability, crash recovery
  - Database tuning as needs evolve

- With the rise of cloud DBs (DB-as-a-Service), the DBA role is eliminated or at least changing drastically in many organizations

Summary

- DBMS used to maintain & query large datasets
- Benefits include
  - Faster app dev (via declarative lang) & easier maintenance (via data independence)
  - recovery from system crashes
  - concurrent access
  - data integrity and
  - data sharing & security
- Levels of abstraction give data independence
- ~~DBAs hold interesting, well-paid jobs~~

Next

- The Entity-Relationship (ER) Model